

Lecture 12 (2026-07-26)

Quick Summary

- **CVP Analysis (Cost-Volume-Profit):** Analyzes the relationship between costs (fixed and variable), volume of output, and resulting profit.
Traditional vs. Contribution Margin Systems:
 - **Traditional:** Focused on Gross Profit (Sales - COGS) and Operating Profit (Gross Profit - Operating Expenses). It does not differentiate between fixed and variable costs.
 - **Contribution Margin:** Focuses on Contribution (Sales - Variable Cost). This system is used to determine how many units must be sold to cover fixed costs.
- **Break-Even Point (BEP):** The point where revenue equals total costs (no profit, no loss). Can be calculated in units or in dollars.
- **Operating Leverage:** Measures the proportion of fixed costs in the total cost structure. Higher leverage indicates higher sensitivity of profit to changes in sales volume.
- **Make or Buy Decision:** Choosing between manufacturing a component/product in-house or purchasing it from an outside supplier.
- **Special Order Decision:** Deciding whether to accept a one-time order at a price different from the standard price.
- **Product Drop Decision:** Deciding whether to continue manufacturing a specific product line or discontinue it based on profitability and other factors.
- **Resource Utilization Decision:** Analyzing how best to use resources available in limited supply (e.g., machine time, labor hours, shelf space).
- **Theory of Constraints:** A method used to identify bottlenecks in production to determine production volume and product mix.
- **Sell or Process Further Decision:** Deciding whether to sell a product in its current state or perform additional processing to increase its selling price.

Key Definitions

- **CVP Analysis:** Cost-Volume-Profit analysis; focuses on how prices, volume, variable costs, fixed costs, and product mix interact to affect profit.
- **Contribution Margin:** The amount remaining from sales revenue after variable costs are deducted (Selling Price - Variable Cost).
- **Contribution Margin per Unit:** The amount contributed by each unit sold toward covering fixed costs and then contributing to profit.
- **Contribution Margin Ratio:** The percentage of sales revenue that is contributed toward fixed costs (Contribution Margin / Net Sales). It remains constant for a given product.
- **Break-Even Point (Units):** The number of units that must be sold to cover all fixed costs.
- **Break-Even Point (Dollars):** The total sales amount required to cover all fixed costs.
- **Multi-Product Break-Even:** Calculation used when a company sells multiple products, requiring a weighted average contribution margin based on the product mix.
- **Cost Structure:** The relative proportion of fixed and variable costs in a company.
- **Operating Leverage:** The ratio of Contribution Margin to Net Income; represents how much profit changes for a given change in sales.
- **Relevant Costs:** Costs that differ between alternatives. Even "fixed" costs can be relevant if they change as a result of a specific decision (e.g., hiring extra staff for a specific order).

- **Avoidable Costs:** Costs that will be eliminated if a specific course of action is taken (e.g., only the specific fixed costs that disappear if a product is dropped).
- **Upstream Costs:** Costs related to the manufacturing of the product (e.g., R&D, product development).
- **Downstream Costs:** Costs related to marketing, distribution, and customer service.
- **Vertical Integration:** When a company is involved in multiple steps of the value chain (from R&D to customer service).
- **Resource Utilization:** Determining how to best allocate limited resources (machine time, labor, shelf space) in the short run.
- **Bottleneck:** A constraint in the production process where capacity is limited.
- **Sell or Process Further:** A decision to determine if additional processing of a product will generate enough additional revenue to cover the additional costs.

Worked Examples

Calculating Units to Cover Fixed Costs - Scenario: 8,000 units sold, fixed costs of 35,000, contribution margin per unit of 3.50. The company currently has a loss of 7,000. - **Problem:** How many more units must be sold to cover fixed costs? - **Step 1:** Calculate total units needed to cover fixed costs: $\$35,000 \div 3.50 = 10,000$ total units. - **Step 2:** Calculate additional units needed: $\$10,000 - 8,000 = 2,000$ additional units. - **Reasoning:** Since the contribution margin is 3.50, each additional unit sold reduces the current loss by 3.50. To find the total units needed to reach a zero-profit/loss point, divide the total fixed costs by the per-unit contribution.

Impact of Unit Changes on Profit/Loss - Single Unit Change: If sales increase by 1 unit, the loss decreases by 3.50 (or profit increases by 3.50 if already profitable). - **100 Unit Change:** $\$100 \times 3.50 = 350$. The loss decreases from 7,000 to 6,650. - **Calculation of Sale Decreases:** If sales decrease by 200 units, the contribution margin decreases by $\$200 \times 12.50 = 2,500$ (Note: The instructor used different figures in this specific example during the lecture, but the logic remains: $\text{\$}\{\text{Change in Units}\} \times \text{\$}\{\text{Contribution Margin per Unit}\} = \text{\$}\{\text{Change in Contribution Margin}\}$).

Strategic Decision Analysis (Happy Daze Game) - Option 1 (Reduce Variable Costs): - Action: Find cheaper raw materials or reduce labor costs (through lower wages or fewer labor hours). - **Calculation:** Original Variable Cost 72,000. Reduced by 10% ($\$72,000 \times 0.90$) = 64,800. - **Outcome:** Reduced variable costs lead to a lower cost base, but must be weighed against quality risks. - **Option 2 (Increase Commission): - Action:** Increase sales commission to drive volume. - **Calculation:** Sales increase by 40,000 (New Total: 140,000). Variable costs increase by 7,200 (New Total: 107,200). - **Reasoning:** Although variable costs increased, the volume increase was much larger, leading to a profit of 200. - **Option 3 (Improve Features and Increase Advertising): - Action:** Higher selling price (due to new features) and increased advertising. - **Calculation:** Sales increase to 148,800; Variable costs to 136,000; Fixed costs increase to 40,000. - **Outcome:** This option yielded the maximum profit (4,800).

Break-Even Point (BEP) Calculations - [shot_00-16-57.png] Break-Even Units Calculation: - Problem: Fixed costs of $\$35,000$, contribution margin per unit of $\$3.50$. - **Calculation:** $\text{\$}\{\text{Break-Even (units)}\} = \text{\$}\{\text{Fixed Costs}\} / \text{\$}\{\text{Contribution Margin Per Unit}\}$ - **Result:** $\$35,000 / 3.50 = 10,000$ units. - **[shot_00-19-12.png] Break-Even Calculations for Multiple Products: - Formula:** $\text{\$}\{\text{Break-Even (Units)}\} = \text{\$}\{\text{Fixed Costs}\} / \text{\$}\{\text{Weighted Average Contribution Margin Per Unit}\}$ - **Multi-Product Selection Calculation (Happy Daze): - Context:** Selling two products (Old Game and New Game). - **Data:** - Old Game: Price $\$12.50$, Variable Cost $\$9.00$, CM $\$3.50$. Volume: 8,000. - New Game: Price $\$15.00$, Variable Cost $\$11.00$, CM $\$4.00$. Volume: 4,500. - **Step 1: Find Proportion of Sales Mix.** - Total Volume = $\$8,000 + 4,500 = 12,500$. - Old Game proportion: $\$8,000 / 12,500 = 64\%$. - New Game proportion: $\$4,500 / 12,500 = 36\%$. - **Step 2: Calculate Weighted Average Contribution Margin.** - $\$(0.64 \times 3.50) + (0.36 \times 4.00) = 2.24 + 1.44 = 3.68$. - **Step 3: Calculate Break-Even.** - $\text{\$}\{\text{BEP (Units)}\} = \text{\$}\{\text{Fixed Costs}\} / 3.68$. - **Reasoning:** We weight the contribution of each product by its share of the total sales volume. As the weighted average CM

increases (by selling more of the high-margin item), the break-even point decreases. - **Target Profit Calculation with Tax Adjustment:** - [shot_00-28-15.png] **Calculation of Before-Tax Profit:** - **Problem:** Target profit after tax is \$100,000. Tax rate is 35%. - **Calculation:** $\$100,000 / (1 - 0.35) = 100,000 / 0.65 = \$153,846$ (approx). - **Logic:** To reach a specific **after-tax** goal, the **before-tax** target must be calculated first. - **Step 2: Calculate Units required to reach target.** -
$$\frac{\text{\text{\{Fixed Costs\}} + \text{\text{\{Before-Tax Profit\}}}}{\text{\text{\{Contribution Margin per Unit\}}}}$$

Special Order Analysis (Sunset Airlines) - Context: Sunset Airlines (regional) offered to provide 150 seats to a corporation at \$125/ticket. - **Data:** - Special Price: \$125. - Normal Price: \$275. - Capacity: 180 seats/plane; 900 total seats across 5 flights. - Current Load: 700. - Available "Excess" Capacity: 200. - Costs: - Food: \$6.50 (Variable). - Fuel, Crew, Maintenance, Depreciation: Fixed. - **Decision Analysis:** - **Step 1 (Capacity):** Since 150 seats are needed and 200 are available, the company has excess capacity to fulfill the order. - **Step 2 (Cost Relevance):** Because fixed costs (fuel, salaries, etc.) are already incurred for the flight regardless of the order, only the variable cost (food) is relevant for the decision. - **Step 3 (Profitability):** Since \$125 (Special Price) > \$6.50 (Variable Cost), the order is profitable even though it is below the normal price. - **Reasoning:** If a company has excess capacity, they can accept a special order even at a lower-than-normal price, provided the price exceeds the incremental (variable) cost of providing that specific service.

Make or Buy Analysis (Birdie Maker) - Context: Birdie Maker produces putters. Ace Putters offers to sell them for \$25 each. - **Data:** - **Make Option:** - Direct Material: \$5,000 - Direct Labor: \$9,000 - Variable Manufacturing OH: \$3,000 - Fixed Manufacturing OH: \$9,500 - **Total Cost to Make:** \$26,500 (or \$26.50 per unit). - **Buy Option:** - Purchase Price: \$25 (per unit). - Fixed Manufacturing OH: \$9,500. - **Total Cost to Buy:** \$34,500 (or \$34.50 per unit). - **Decision Analysis:** - **Step 1: Comparison.** Comparing "Make" vs. "Buy" results in a \$9,000 difference (or \$9.00 per unit). - **Step 2: Logic of Fixed Costs.** The instructor noted that because the \$9,500 fixed cost is incurred regardless of the decision, it can be either added to both options or removed from both. The resulting comparison remains the same. - **Step 3: Conclusion.** In this case, the purchase price of \$25 is higher than the variable cost of making the item (\$17). Thus, manufacturing the item is cheaper. - **Reasoning:** When deciding whether to make or buy, only the costs that **change** based on the decision (variable costs) should be the primary focus of the difference, as fixed costs are "sunk" unless they change specifically due to the choice.

Drop Product Analysis (Mud and Snow Tires) - Context: Deciding whether to keep or drop "Mud and Snow" (M&S) tires. - **Data:** - **Current State (Both Products):** - Sales: \$175,000 (150k Other + 25k M&S) - Material: \$62,600 - Labor: \$35,000 - Variable OH: \$16,000 - Fixed OH: \$28,000 (21k Other + 7k M&S) - **Proposed State (Drop M&S):** - Sales: \$150,000 - Material: \$50,600 - Labor: \$30,000 - Variable OH: \$14,000 - Fixed OH: \$26,000 (Only \$2,000 of the M&S fixed cost is avoidable). - **Decision Analysis:** - **Step 1: Compare Net Income.** - Profit with M&S: \$33,400 (Calculation: \$175k - 62.6k - 35k - 16k - 28k\$). - Profit without M&S: \$31,400 (Calculation: \$150k - 50.6k - 30k - 14k - 26k\$). - **Step 2: Conclusion.** Since the profit is higher when keeping the product, the company should keep the Mud and Snow line. - **Reasoning:** The key is to identify **avoidable** fixed costs. If the profit with the product is higher than without it, the product should be kept. Furthermore, qualitative factors (like the diversity of choice for customers) must be considered.

Resource Utilization (Birdie Maker Golf Balls) - Scenario: Birdie Maker has limited machine time. They must decide how to allocate production between Pro models and Tour models. - **Data:** - **Pro Model:** Price = \$150, Contribution = \$150, Time required = 30 mins (0.5 hours). - **Tour Model:** Price = \$150, Contribution = \$150, Time required = 55 mins. - **Calculations:** - **Pro Contribution per machine hour:** $\$150 \div 0.5 = 300$ - **Tour Contribution per machine hour:** Approximately 200. - **Decision Analysis:** - **Selection:** The Pro model should be prioritized for production because it yields a higher contribution per unit of the constrained resource (machine hour). - **Multi-Product Strategy:** If production is limited, prioritize the product with the higher contribution per hour. If demand for both is limited, fulfill the demand for the higher-profit-per-hour item first, then use the remaining "idle" time for the other item. - **Reasoning:** When a resource is in limited supply (short-run), the decision should be based on the **Contribution per unit of the constrained resource** (e.g., CM per machine hour) rather than just the total contribution per unit. A product that looks "premium" (higher price) may be less

desirable if it consumes disproportionately more of a scarce resource.

Sell or Process Further (Unfinished Furniture) - Scenario: Decision to sell furniture as unfinished or process it further into finished furniture. - **Data:** - **Current State (Unfinished):** Selling Price = \\$150. - **Processing Step:** Additional Cost = \\$45. - **New State (Finished):** Selling Price = \\$225 (Price increases by \\$75). - **Decision Analysis:** - **Step 1: Calculate Incremental Profit.** - Additional Revenue: \\$75 - Additional Cost: \\$45 - Net Incremental Gain: \\$30 - **Conclusion:** Sell as "finished" because the additional revenue exceeds the additional cost. - **Reasoning:** When deciding whether to process further, only the **incremental** costs and revenues resulting from the additional step should be compared. Fixed costs incurred before this decision are irrelevant.

Formulas & Rules

- **Contribution:** $\text{Selling Price} - \text{Variable Cost}$
- **Contribution Margin Ratio:** $\frac{\text{Contribution Margin}}{\text{Net Sales}} \times 100$ (or $\frac{\text{Selling Price} - \text{Variable Cost}}{\text{Selling Price}}$)
- **Break-Even Point (Units):** $\frac{\text{Fixed Costs}}{\text{Contribution Margin per Unit}}$
- **Break-Even Point (Dollars/Value):** $\frac{\text{Fixed Costs}}{\text{Contribution Margin Ratio}}$
- **Impact of Sales Change:** $\text{Change in Sales} \times \text{Contribution Margin Ratio} = \text{Change in Contribution Margin}$
- **Weighted Average Contribution Margin:** $\sum (\text{CM per Unit} \times \text{Sales Mix \%})$
- **Target Profit (Units):** $\frac{\text{Fixed Costs} + \text{Target Profit (Before Tax)}}{\text{Contribution Margin per Unit}}$
- **Before-Tax Profit Calculation:** $\text{After-Tax Profit} / (1 - \text{Tax Rate})$
- **Operating Leverage:** $\frac{\text{Contribution Margin}}{\text{Net Income}}$
- **Special Order Accept/Reject:** Accept if $\text{Special Price} > \text{Variable Cost}$ AND there is excess capacity.
- **Make vs. Buy Comparison:** Compare $\text{Variable Cost to Make}$ vs. Purchase Price . Fixed costs are ignored unless they are specifically "avoidable" or "incremental."
- **Resource Ranking:** When resources are limited, rank products by $\frac{\text{Contribution per Unit}}{\text{Units of Constrained Resource}}$.
- **Sell or Process Further:** Compare $\text{Incremental Revenue}$ vs. Incremental Cost .

Tricky Logic & Traps

- **Traditional vs. Contribution Reporting:** The exam may ask you to distinguish between the two. **Trap:** Remember that while both systems result in the same Net Income, the Traditional system does not distinguish between fixed and variable costs, whereas the Contribution system does.
- **Strategic Decision Assessment (Qualitative vs. Quantitative):** For Special Orders, Make vs. Buy, and Drop Decisions, always consider both factors. A deal might be profitable (quantitative) but damage brand loyalty or employee morale (qualitative).
- **Fixed Costs and Relevance: Trap:** Fixed costs are not always irrelevant. They become relevant if they change due to the decision (e.g., needing a new pilot or extra supervisor for a special order).
- **Calculation of Break-Even in Dollars:** To find BEP in dollars, you must use the **Contribution Margin Ratio**. Using the "per unit" figure for a dollar calculation is a common error.
- **Contribution Margin as a Constant:** The Contribution Margin Ratio is treated as a constant percentage of sales. This is the basis for calculating how much profit/loss will change based on a percentage change in volume.
- **Incremental Fixed Costs:** When making a new decision (e.g., launching a new game), only include the **incremental** fixed costs (the costs that will specifically occur because of this decision) rather than total existing fixed costs.

- **Impact of Sales Mix:** If the sales mix shifts toward a product with a higher Contribution Margin, the weighted average CM increases and the Break-Even Point decreases. Conversely, if the mix shifts toward a lower CM product, the BEP increases.
- **Operating Leverage and Sensitivity:** A higher operating leverage means a higher proportion of fixed costs in the cost structure. **Trap:** Higher operating leverage results in **higher sensitivity** to changes in revenue. Even a small percentage change in sales can lead to a large percentage change in profit.
- **Taxes in Target Profit:** When a target profit is given as "after tax," it must be converted to "before tax" using the formula $\frac{\text{After-tax Profit}}{1 - \text{Tax Rate}}$ before being added to fixed costs in the calculation.
- **Avoidable Costs in Drop Decisions:** When evaluating if a product should be dropped, only the **avoidable** fixed costs should be deducted from the "without" scenario. Unhandled (common) fixed costs remain regardless of the decision.
- **The "Bundle" Trap:** Just because a product (like a pillow cover) has a lower margin or isn't "premium" doesn't mean it should be dropped. **Trap:** Customers often expect a "full range" from a single seller; removing a lower-margin item can alienate customers who then won't purchase the higher-margin items.
- **Short-run vs. Long-run Constraints:** In the short-run, you must manage existing limitations (e.g., limited machine hours). In the long-run, you can expand capacity (e.g., buying more machines or hiring more labor).
- **Ranking by Constraint: Trap:** When multiple products compete for a limited resource, you must rank them by **Contribution per Unit of the Constraint** (e.g., CM per machine hour) rather than just the standard Contribution Margin per unit.
- **Sell or Process Further Logic:** When deciding whether to process further, the only relevant factor is whether the **incremental** revenue covers the **incremental** cost. Historical fixed costs are irrelevant to this specific decision.

Exam Pointers

- **Calculation of "Extra" Units:** If the question asks how many **more** units must be sold to reach break-even, subtract the current units sold from the total break-even units.
- **Identifying the "Best" Option:** When comparing multiple options for a company (like Happy Daze), the "best" option is the one that results in the highest profit while acknowledging the qualitative risks of each.
- **Breakeven Graph:** Remember the visual: Break-even is the point where the **Revenue** line and the **Total Cost** line intersect.
- **High Fixed Cost Industries:** Be prepared to identify industries with high operating leverage (e.g., hotels, hospitals, airlines) because their cost structures are dominated by fixed costs, making their profitability highly sensitive to volume.
- **Special Order Eligibility:** To accept a special order, verify: 1. It is below normal price. 2. There is excess capacity. 3. Price exceeds variable costs.
- **Make vs. Buy Calculation:** To compare, only look at the costs that **change** between the two choices (the variable components). Adding the same fixed cost to both sides does not change the final decision.
- **Decision Priority:** When resource-constrained, priority is given to the product with the highest contribution per unit of the constrained resource.

Sources

- PV ratio practice question.docx
- [shot_00-50-42.png] Excess Capacity
- [shot_00-51-39.png] Special Order Example

- [shot_00-56-07.png] Fixed Costs Can Be Relevant
- [shot_00-57-06.png] Relevant Costs Analysis
- [shot_00-59-31.png] Vertical Integration and Value Chain
- [shot_01-07-50.png] Qualitative Factors - Product Mix Impact
- Make or Buy and Special Order.docx
- [shot_01-08-58.png] Resource Utilization Decisions
- [shot_01-15-00.png] Sell or Process Further